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Trading-day Variations Multiple Regression Models with Random Parameters

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Summary

A large class of flows and stocks series related to production, shipments, sales and inventories are affected by trading-day or calendar variations. Trading-day variations represent the 'within-month variations' due to the number of times a particular day or days of the week occur in a calendar month. These variations are systematic and may strongly influence month-to-month comparisons.

A simple trading-day regression model, developed by Young (1965), is currently used by the X-11-ARIMA (Dagum, 1980) and the Census Method II-X-11 variant (Shiskin, Young & Musgrave, 1967) to estimate trading-day variations. This simple model assumes that the trading-day coefficients are fixed and estimated by ordinary least squares. As long as the relative weight of daily activities is constant throughout the length of the chosen span of the series, this deterministic model produces reasonable estimates. However, that is not always a realistic assumption.

The purpose of this paper is to introduce a multiple regression model that allows for stochastic changes of the trading-day coefficients used to calculate trading-day variations. Estimation and statistical inference for the trading-day regression are thoroughly discussed. Examples of the stochastic and deterministic models are given for real data.

Key words: Fibonacci line search; Fixed interval smoother; Kalman filter; Maximum likelihood estimation; Method of scoring; Rao score test; Spectral densities.

I Introduction

The trading-day regression model developed by Young (1965) is currently used by the X-11-ARIMA (Dagum, 1980) and the Census Method II-X-11 variant (Shiskin, Young & Musgrave, 1967) to estimate the trading-day component. The same model for trading-day variations has been discussed by Cleveland & Devlin (1980 and 1982), Hillmer (1982), Cleveland & Grupe (1983), Hillmer, Bell & Tiao (1983), Bell & Hillmer (1983), Kitagawa & Gersch (1984) and Salinas & Hillmer (1987). This model, called here 'deterministic', assumes that the trading-day coefficients remain constant throughout the chosen span of the series. For some socio-economic time series this assumption might be too restrictive and a trading-day multiple regression model with stochastic trading-day coefficients may be more adequate. This is particularly true for series affected by changes in store opening hours which influence consumer shopping patterns and by more retail outlets open on Sundays. Simple stochastic models for the trading-day coefficients, namely, a random walk and a random walk with a random drift were proposed by Monsell (1983) and Dagum & Quenneville (1988).

In this paper we extend previous works by introducing (1) a general stochastic model for the trading-day coefficients, the parameters of which are estimated by maximum

likelihood, and (2) a test to assess whether the trading-day variations are deterministic or stochastic. Section 2 gives a definition of trading-day variations. Section 3 discusses the proposed stochastic model together with the deterministic model. Section 4 deals with the estimation procedure of the stochastic model. Section 5 discusses inference problems. In Section 6, the stochastic and deterministic models are applied to two real series affected by trading-day variations. Finally, Section 7 gives the conclusions of this study.

2 Definition of Trading-Day Variations

Consider a flow series where the monthly total y_t is the sum of the daily total $y_{d,t}$ $d = 1, 2, \dots, N_t$, where $N_t = 28, 29, 30$ or 31 is the number of days in the month. For example, in a month of 30 days starting on a Monday,

$$\begin{aligned} y_t &= y_{1,t} + y_{2,t} + y_{3,t} + \dots + y_{7,t} \\ &\quad + y_{8,t} + y_{9,t} + y_{10,t} + \dots + y_{14,t} \\ &\quad + y_{15,t} + y_{16,t} + y_{17,t} + \dots + y_{21,t} \\ &\quad + y_{22,t} + y_{23,t} + y_{24,t} + \dots + y_{28,t} \\ &\quad + y_{29,t} + y_{30,t} \\ &= \sum_{i=1}^{30} y_{i,t}. \end{aligned} \quad (2.1)$$

Equation (2.1) can be written as,

$$\begin{aligned} y_t &= \frac{5}{5} (y_{1,t} + y_{8,t} + y_{15,t} + y_{22,t} + y_{29,t}) \\ &\quad + \frac{5}{5} (y_{2,t} + y_{9,t} + y_{16,t} + y_{23,t} + y_{30,t}) \\ &\quad + \frac{4}{4} (y_{3,t} + y_{10,t} + y_{17,t} + y_{24,t}) \\ &\quad \dots \\ &\quad + \frac{4}{4} (y_{7,t} + y_{14,t} + y_{21,t} + y_{28,t}) \end{aligned} \quad (2.2)$$

or

$$y_t = 5\bar{y}_{1t} + 5\bar{y}_{2t} + 4\bar{y}_{3t} + 4\bar{y}_{4t} + 4\bar{y}_{5t} + 4\bar{y}_{6t} + 4\bar{y}_{7t}, \quad (2.3)$$

where the $\bar{y}_{it}$'s represent the mean of the day $i = \text{Monday (1), Tuesday (2), } \dots, \text{ Sunday (7)}$, and are called the daily effects. Let N_{it} be the corresponding number of Mondays (N_{1t}), Tuesdays (N_{2t}), $\dots$, and Sundays (N_{7t}) in a month t . Equation (2.3) is therefore written as,

$$y_t = \sum_{i=1}^7 N_{it} \bar{y}_{it}. \quad (2.4)$$

Let

$$\bar{y}_{0t} = \frac{1}{7} \sum_{i=1}^7 \bar{y}_{it}$$

be the average of the daily effects. (Note that $\bar{y}_{0t}$ is not the monthly average $\bar{y}_t$). Then equation (2.4) becomes:

$$y_t = \sum_{i=1}^6 (N_{it} - N_{7t})(\bar{y}_{it} - \bar{y}_{0t}) + N_{7t} \bar{y}_{0t}. \quad (2.5)$$

In this study we define the trading-day effect or variations in month t as the first term of the right hand side of equation (2.5). That is,

$$D_t = \sum_{i=1}^6 (N_{it} - N_{7t})(\bar{y}_{it} - \bar{y}_{0t}). \quad (2.6a)$$

The trading-day effect represents the effect attributed to the number of times each day of the week occurs in month t . There is no trading-day effect if all the $\bar{y}_{it}$'s are equal. Similarly, for the month of February, except in leap year, $N_{it} = N_{7t}$ $i = 1, \dots, 6$, and there is no trading-day effect. For notational convenience (2.6a) can be expressed as,

$$D_t = \sum_{i=1}^6 x_{it} \delta_{it} = x'_t \delta_t, \quad (2.6b)$$

where $\delta_{it} = (\bar{y}_{it} - \bar{y}_{0t})$ $i = 1, \dots, 6$ and $x_{it} = N_{it} - N_{7t}$. The δ_{it} 's, called the trading-day coefficients, represent the difference between the Monday, Tuesday, $\dots$, and Saturday daily effects $\bar{y}_{it}$ and their average $\bar{y}_{0t}$, for month t . The Sunday trading-day coefficient is,

$$\delta_{7t} = \bar{y}_{7t} - \bar{y}_{0t} = - \sum_{i=1}^6 \delta_{it}. \quad (2.7)$$

The x_{it} is an indicator that takes the values -1 , 0 or 1 . If there are 4 Sundays ($N_{7t} = 4$) in a month then, $x_{it} = 1$ for those days that occur 5 times and equal to zero for those days that occur 4 times in that month. If there are 5 Sundays then, $x_{it} = -1$ for those days that occur 4 times and equal to zero for those days that occur 5 times. Hence when there are 4 Sundays the trading-day effect (2.6) is the sum of the trading-day coefficients (δ_{it}) of the extra days. When there are 5 Sundays, the trading-day effect is the negative sum of the δ_{it} 's of the days that occur 4 times in the month. But the negative sum of the δ_{it} 's of the days that occur 4 times is exactly equal to the sum of the δ_{it} 's of the days that occur 5 times in the month. This is because

$$\sum_{i=1}^7 \delta_{it} = \sum_{i=1}^7 (\bar{y}_{it} - \bar{y}_{0t}) = 0,$$

by (2.7). Hence, the trading-day effect is always the sum of the trading-day coefficients of the days that occur 5 times in the month.

3 Deterministic and Stochastic Models for Trading-day Coefficients

The estimation of trading-day variations in X-11-ARIMA and Census Method II X-11 variant is done sequentially after other variations, namely, seasonality and trend-cycle are removed from the data. Other model-based seasonal adjustment methods estimate trading-day variations simultaneously with the remaining components via an iterative process where the seasonality and the trend-cycle are initially assumed to follow very simple stochastic models. Hence, given the time series, $\{y_t\}$, we assume that the trend-cycle and seasonal variations have been removed (if present) and denote the new series as:

$$w_t = D_t + e_t, \quad t = 1, \dots, T \quad (3.1)$$

where $e_t \sim \text{NID}(0, \sigma^2)$ and T is the number of observations. The normality assumption is only necessary for inferential purposes. If the normality assumption is dropped least squares estimates are obtained instead of maximum likelihood estimates.

The trading-day regression model with fixed parameters developed by Young (1965) assumes that $\delta_{it} = \delta_i$ for all t . In this case, the deterministic trading-day effect is,

$$D_t^d = \sum_{i=1}^6 x_{it} \delta_i \quad (3.2)$$

and

$$w_t = \sum_{i=1}^6 x_{it} \delta_i + e_t, \quad (3.3)$$

where the δ_i 's denote fixed unknown trading-day coefficients and are estimated by ordinary least squares (OLS). The deterministic model assumes that the trading-day coefficients remain constant throughout the chosen span of the series. This assumption might be too restrictive for some series where consumer shopping patterns may change randomly through time. To accommodate such changes, we consider stochastic trading-day effects, and express (3.2) and (3.3) as,

$$D_t^s = \sum_{i=1}^6 x_{it} \delta_{it} \quad (3.4)$$

and

$$w_t = \sum_{i=1}^6 x_{it} \delta_{it} + e_t, \quad (3.5)$$

respectively. The model for the trading-day coefficients is assumed to satisfy:

$$(1 - B)^k \underline{\delta}_t = \underline{\chi}_t \quad (3.6)$$

where

$$\underline{\delta}_t = (\delta_{1t}, \dots, \delta_{6t})'$$

is the vector of trading-day coefficients in month t , and,

$$\begin{aligned} \underline{\chi}_t &= (\chi_{1t}, \dots, \chi_{6t})' \\ \underline{\chi}_t &\sim \text{NID}(\underline{0}_6, \sigma_\chi^2 \mathbf{I}_6), \end{aligned} \quad (3.7)$$

is the noise vector. B denotes the backshift operator ($B\underline{\delta}_t = \underline{\delta}_{t-1}$) and $\mathbf{I}_6$ is the identity matrix of order 6.

The rational behind model (3.6) is to adjust locally a polynomial of degree k to the trading-day coefficients. The degree k of the difference can be selected using Akaike's criterion discussed in Section 5. Statistical tests to distinguish between deterministic and stochastic models for the trading-day coefficients are also discussed in Section 5.

For a relative large sample of trade series (retail, wholesale and international) we found that the random walk model (the stochastic model (3.6) with $k = 1$), or the deterministic trading-day regression gave a good fit. The value of $k = 1$ allows the trading-day coefficients to change smoothly and rapidly enough. For $k = 2$, (to be called here the quadratic model) equation (3.6) puts a more rigid structure on the $\underline{\delta}_t$'s and we found that it takes more time for the trading-day coefficients to adapt to their new level. This will be illustrated in Section 6.

One problem in multiple regression model with random parameters is to estimate the hyper-parameter σ_χ^2 or the ratio σ_χ^2/σ^2 . For $k = 1$, Monsell (1983) uses a simple search over a pre-specified set of numbers. This author, however is not fully satisfied with his approach and stresses the need for pursuing further research to improve the estimation procedure. A first step on this direction is given by Dagum & Quenneville (1988) where the search over a grid is replaced by the Fibonacci line search. In this study the maximum likelihood estimators (MLE) of the hyper-parameter σ_χ^2/σ^2 is obtained by the method of

scoring. Under suitable regularity conditions this method not only provides a point estimate but also gives an estimate of the asymptotic variance of the estimator. This is a major improvement over the previous methods.

4 Estimation of the Stochastic Model for Trading-day

The stochastic model for trading-day regression is written in state space form and the estimates of the trading-day coefficients δ_t , $t = 1, \dots, T$, together with their mean squared error matrices are obtained with the Kalman filter and the fixed interval smoother.

4.1 State Space Representation

The Kalman filter is a set of recursive equations used to update and smooth estimates of current and previous state vectors when new data become available and also to predict future values. (A detailed discussion can be found in Anderson & Moore, 1979).

The state space model consists of a measurement equation, namely,

$$w_t = \underline{z}_t' \underline{\alpha}_t + \epsilon_t, \quad t = 1, \dots, T \quad (4.1)$$

and a transition equation, namely,

$$\underline{\alpha}_t = \mathbf{G} \underline{\alpha}_{t-1} + \underline{\eta}_t, \quad t = 1, \dots, T \quad (4.2)$$

where $\underline{\alpha}_t$ is an $m \times 1$ state vector, $\underline{z}_t$ is a $m \times 1$ fixed vector, $\mathbf{G}$ is a fixed $m \times m$ matrix and the errors ϵ_t and η_t are independent. It is further assumed that $\epsilon_t \sim \text{NID}(0, \sigma^2)$ and $\eta_t \sim \text{NID}(0, \sigma^2 \mathbf{Q})$ where $\mathbf{Q}$ is a fixed $m \times m$ matrix and σ^2 is a scalar. Although $\mathbf{Q}$ may depend on unknown parameters it is regarded as being fixed and known for the purpose of the Kalman filter.

We denote by $\underline{a}_{t-1}$ the minimum mean squared estimator (MMSE) of $\underline{\alpha}_{t-1}$ based on all the information up to and including $t-1$, and by $\sigma^2 \mathbf{P}_{t-1}$ the MSE matrix of $\underline{a}_{t-1}$, i.e., the covariance matrix of $\underline{a}_{t-1} - \underline{\alpha}_{t-1}$. The MMSE of $\underline{\alpha}_t$, given $\underline{a}_{t-1}$ and $\sigma^2 \mathbf{P}_{t-1}$ is denoted by $\underline{a}_{t|t-1}$ with MSE matrix $\sigma^2 \mathbf{P}_{t|t-1}$. Once w_t becomes available, this estimator of $\underline{\alpha}_t$ is updated and denoted by $\underline{a}_t$ with MSE matrix $\sigma^2 \mathbf{P}_t$. Starting values $\underline{a}_0$ and $\mathbf{P}_0$ are needed to implement the Kalman filter.

The Kalman filter yields the MMSE of $\underline{\alpha}_t$, given the information available up to time t . However, once all the observations are available, a better estimator can be obtained. One of the techniques for computing such estimators is the fixed interval smoother. The fixed interval smoother is a set of recursions which start with the Kalman filter estimates $\underline{a}_T$ and $\mathbf{P}_T$, and works backwards. We denote by $\underline{a}_{t|T}$ and $\sigma^2 \mathbf{P}_{t|T}$, the smoothed estimate and its covariance matrix.

A convenient state space representation of the random walk model ($k=1$) for the trading-day coefficients (3.6) along with the observation equation (3.5) is obtained through the following equivalences with the transition equation (4.2) and the measurement equation (4.1):

$$\underline{\alpha}_t = \delta_t, \quad \underline{z}_t' = (x_{1t}, \dots, x_{6t}), \quad \epsilon_t = e_t, \quad (4.3a)$$

$$\mathbf{G} = \mathbf{I}_6, \quad \underline{\eta}_t = \chi_t \quad \text{and} \quad \mathbf{Q} = \frac{\sigma_\chi^2}{\sigma^2} \mathbf{I}_6. \quad (4.3b)$$

For the quadratic model ($k = 2$), the equivalences are given by:

$$\alpha'_t = [\delta'_t, \delta'_{t-1}], \quad z'_t = (x_{1t}, \dots, x_{6t}, 0'_6), \quad \epsilon_t = e_t, \quad (4.4a)$$

$$\mathbf{G} = \begin{bmatrix} 2\mathbf{I}_6 & -\mathbf{I}_6 \\ \mathbf{I}_6 & 0_6 \end{bmatrix}, \quad \eta_t = \begin{bmatrix} z_t \\ 0_6 \end{bmatrix} \quad \text{and} \quad \mathbf{Q} = \begin{bmatrix} \frac{\sigma_x^2}{\sigma^2} \mathbf{I}_6 & 0_6 \\ 0_6 & 0_6 \end{bmatrix} \quad (4.4b)$$

It is clear from (4.3b) and (4.4b) that $\mathbf{Q}$ depends on the unknown parameters σ_x^2/σ^2 . The estimation of this parameter and σ^2 as well as the estimation of the initial conditions α_0 and $\mathbf{P}_0$ are discussed in the next sections.

4.2 Estimation of the Initial Conditions

For a stationary process, the theoretical mean value and covariance matrix of the state vector can be used to initiate the Kalman filter. However, for a nonstationary process the theoretical mean and covariance matrix cannot be defined. In order to obtain initial estimates of α_0 and $\mathbf{P}_0$ we apply the Kalman filter over the time-reversed series with the zero vector as the initial state vector of the reversed series and its MSE matrix equal to the identity matrix multiplied by a large constant. Estimates of the initial state vector of the original series and its MSE matrix are obtained by forecasting. The hyper-parameters of the reversed series are computed using the Fibonacci line search method mentioned in the next section.

4.3 Estimation of the Parameters σ_x^2/σ^2 and σ^2

Maximum Likelihood Estimators (MLE) are used for the estimation of the parameters σ_x^2/σ^2 and σ^2 . Using the prediction error decomposition (Harvey 1981), the likelihood function L , can be written as follows;

$$-2 \log L = T \log 2\pi + T \log \sigma^2 + \sum_{t=1}^T \log f_t + \frac{1}{\sigma^2} \sum_{t=1}^T \frac{v_t^2}{f_t} \quad (4.5)$$

where T is the number of observations and v_t and f_t are defined by,

$$v_t = w_t - z'_t a_{t|t-1},$$

$$f_t = z'_t \mathbf{P}_{t|t-1} z_t + 1.$$

Differentiation of (4.5) with respect to σ leads to $\hat{\sigma}^2$, the MLE of σ^2 , given by:

$$\hat{\sigma}^2 = \frac{1}{T} \sum_{t=1}^T \frac{v_t^2}{f_t}. \quad (4.6)$$

The scalar parameter, σ^2 , may be concentrated out of the log-likelihood function leaving the concentrated log-likelihood function,

$$-2 \log L_c = T \log 2\pi + T + T \log \hat{\sigma}^2 + \sum_{t=1}^T \log f_t.$$

Numerical optimization has to be carried out with respect to the remaining parameter σ_x^2/σ^2 to find its MLE. The Fibonacci line search method (Bazaraa & Shetty, 1979) is used to find the maximum between zero and one. This assumes that the noise in the signal α_t is less than the noise in the measurements w_t . This estimate is further improved using

the method of scoring. Let $x = \sigma_x^2/\sigma^2$ be the unknown parameter to be estimated. The method of scoring is the iterative scheme:

$$x^{(i)} = x^{(i-1)} + I^{-1}(x^{(i-1)})D \log L(x^{(i-1)}),$$

where $x^{(i)}$ is the estimate of x at the i -th iteration, $I(x)$ is the Fisher information number evaluated at x , $D \log L(x)$ is the derivative of the log-likelihood function evaluated at x and $x^{(0)}$ is the estimated parameter σ_x^2/σ^2 obtained from the Fibonacci line search.

5 Inference

The general model for trading-day variations is,

$$w_t = x_t' \delta_t + e_t, \quad (5.1)$$

$$(1 - B)^k \delta_t = \chi_t, \quad \delta_0 \text{ unknown} \quad (5.2)$$

with $e_t \sim \text{NID}(0, \sigma^2)$, σ^2 unknown and $\chi_t \sim \text{NID}(0_6, \sigma_x^2 \mathbf{I}_6)$, $\sigma_x^2 > 0$ is unknown for the stochastic model and $\sigma_x^2 = 0$ for the deterministic model. Under the deterministic model the standard F -test can be used to test the significance of the deterministic trading-day coefficients. This section focuses on the inferential problems that involve the stochastic model for the trading-day coefficients.

5.1 Inference on the Degree k of Difference

The degree k of the difference adequate for each series can be selected using Akaike's criterion (AIC) as discussed in Kitagawa & Gersch (1984). The AIC for a particular k is:

$$\text{AIC}(k) = -2 \log(\text{maximized likelihood}) + 2(6k + 1). \quad (5.3)$$

The value of k for which $\text{AIC}(k)$ is smallest is the best model. Here the number $6k$ refers to the dimension of the initial state vector and 1 is for the parameter σ_x^2/σ^2 , both having to be estimated.

Another possible criterion to select the best model is the Schwarz criterion defined by:

$$S(k) = -2 \log(\text{maximized likelihood}) + (6k + 1) \log n. \quad (5.4)$$

As pointed out by Schwarz (1978) the $S(k)$ criterion leans more than $\text{AIC}(k)$ towards lower-dimensional models. Moreover, the $S(k)$ criterion is asymptotically optimal.

5.2 Inference on the Variance σ_x^2

To distinguish between the deterministic and the random walk models for trading-day coefficients a test of a composite hypothesis has to be done. For the random walk model:

$$\begin{aligned} \delta_t &= \delta_{t-1} + \chi_t \\ &= \delta_{t-2} + \chi_{t-1} + \chi_t \\ &= \delta_0 + \chi_1 + \dots + \chi_t. \end{aligned} \quad (5.5)$$

Using (5.1), we deduce,

$$w_t = x_t' \delta_0 + x_t' \sum_{j=1}^t \chi_j + e_t. \quad (5.6)$$

Hence it follows:

$$\mathcal{E}(w_t) = x_t' \delta_0,$$

and

$$V(w_t) = t x_t' x_t \sigma_x^2 + \sigma^2,$$

$$\text{Cov}(w_t, w_s) = \min(s, t) x_s' x_s \sigma_x^2.$$

Let $\mathbf{X} = (x_1, \dots, x_T)'$, $W = (\min(s, t))$ and $*$ denote the elementwise product, then:

$$w \sim N(\mathbf{X} \delta_0, \sigma^2 \mathbf{I}_T + \sigma_x^2 \mathbf{W}^* (\mathbf{X} \mathbf{X}')). \quad (5.7)$$

Let $\mathbf{V} = \mathbf{W}^* (\mathbf{X} \mathbf{X}')$. The hypothesis to be tested is: $H_0: \sigma_x^2 = 0$, σ^2 and δ_0 unknown, against $H_1: \sigma_x^2 > 0$, σ^2 and δ_0 unknown. H_0 corresponds to the specification of the deterministic model and H_1 to that of the random walk model for the trading-day coefficients.

Franzini & Harvey (1983) and Nyblom (1986) developed most powerful tests based on the likelihood ratio criterion. But their test statistics require specific values of σ_x^2 under the alternative H_1 . Locally most powerful tests are also available (see for example: Nyblom & Mäkeläinen (1983) and Nabeya & Tanaka (1988)) to test similar hypotheses but their test statistics are cumbersome to compute. We discuss Rao's score test (Rao (1973)) which has, asymptotically, a chi-square distribution and is easy to compute.

The hypothesis to be tested is invariant in translation $w \rightarrow w + X \delta$ where δ is a (6×1) vector. Let the columns of $\mathbf{P}$ form an orthonormal basis for the orthogonal complement of the columns space of $\mathbf{X}$ and consider the transformation:

$$z = \mathbf{P}' w. \quad (5.8)$$

It follows from (5.7) that;

$$z \sim N(0_{T-6}, \sigma^2 \mathbf{I}_{T-6} + \sigma_x^2 \mathbf{P}' \mathbf{V} \mathbf{P}). \quad (5.9)$$

Let $\Sigma = \sigma^2 \mathbf{I}_{T-6} + \sigma_x^2 \mathbf{P}' \mathbf{V} \mathbf{P}$. Our hypothesis is now, $H_0: \sigma_x^2 = 0$, σ^2 unknown against $H_1: \sigma_x^2$ and σ^2 unknown.

As shown in the Appendix the Rao Score Test statistic for this hypothesis is,

$$S = \frac{(\hat{e}' \mathbf{V} \hat{e} - \hat{\sigma}_0^2 \text{trace}(\mathbf{M} \mathbf{V}))^2}{2 \hat{\sigma}_0^4 \text{trace}(\mathbf{M} \mathbf{V} \mathbf{M} \mathbf{V})}, \quad (5.10)$$

where $\mathbf{M} = \mathbf{I}_T - (\mathbf{X}(\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}')$, $\hat{e} = \mathbf{M} w$ is the residual vector from the deterministic regression and:

$$\hat{\sigma}_0^2 = \frac{1}{T-6} \hat{e}' \hat{e} \quad (5.11)$$

is the estimate of σ^2 from the deterministic regression. The test called for the rejection of the null hypothesis at level α if the observed value of S is greater than the upper α -percentile point of a chi-square distribution with one degree of freedom.

6. Application

The deterministic and the two stochastic models for the estimation of the trading-day coefficients were applied on a large sample of real series affected by trading-day variations. The series were from the sectors of Retail Trade, Wholesale Trade, and International Trade.

All the series were first seasonally adjusted using the X-11-ARIMA method without ARIMA extrapolations and with the default options. Therefore, the input series w_t is a

Table 1A
Total retail trade sales—all stores—Nova Scotia (I). Transformed residuals series¹

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
77	161	-221	-300	574	-641	-363	480	-583	447	-223	342	693
78	-883	-34	243	-559	-11	558	-182	-99	585	-503	-155	-36
79	272	-428	935	-1113	171	1062	-921	1344	-1287	-230	1136	-954
80	44	712	-756	-136	1212	-903	-62	488	-275	696	-333	-1167
81	720	499	-657	695	409	-651	233	11	-817	514	-239	490
82	84	-443	-230	1251	-1097	0	682	-732	293	-168	-177	874
83	-563	-796	1613	-640	-1307	1451	73	-513	639	-739	-631	958
84	-865	638	407	-552	330	795	-1350	383	-206	-337	910	-928
85	693	-288	285	186	128	-475	-139	1008	-865	30	530	-1289
86	1514	161	-1202	705	608	-438	-184	-788	393	777	-640	-194

(1) Entries have to be multiplied by 10^{-3} .

Table 1B
Total retail trade sales—department stores—Canada (II). Transformed residuals series¹

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
77	-225	270	-243	871	-721	-765	234	288	490	-562	309	-173
78	-338	-65	810	-315	81	-31	-827	-202	1473	-652	78	217
79	-396	-731	1833	-1233	168	119	-655	1230	-598	-12	102	-585
80	962	470	-780	-440	1181	-1680	938	-8	170	248	-600	-565
81	1955	-437	-1076	352	-358	611	-149	-273	-317	1014	-362	298
82	-315	105	-569	603	-273	-642	623	-270	421	-370	-157	481
83	-240	-24	1476	-1735	-1455	1630	104	8	215	-353	-649	886
84	-719	897	63	-941	-27	1143	-1006	-222	293	-213	830	-514
85	118	-504	537	257	528	-1401	-090	1451	-995	-116	756	-676
86	738	-603	-87	492	1390	-1917	453	100	-191	436	-405	214

(1) Entries have to be multiplied by 10^{-3} .

series of residual (I_t) obtained from Table B-13 and transformed by the following equation:

$$w_t = (I_t/100 - 1)N_t \tag{6.1}$$

where N_t is the number of days in month t .

Table 2
Summary of the estimation procedure for σ_x^2/σ^2 .
(I) (II)

Step ⁽¹⁾	$k = 1$	$k = 2$	$k = 1$	$k = 2$
Step1	0.0153	0.00031	0.00031	0.00031
Step3	0.0103	0.000046	0.00031	0.000046
Step4	0.0103	0.000026	NA ⁽²⁾	0.000043
AIC ⁽³⁾	-211	-170	-179	-132
S ⁽⁴⁾	10.0933		0.1400	
SL ⁽⁵⁾	0.0015		0.7083	

(1) Refer to Section 6 for definition.
(2) Non-available: Method of scoring leads to a negative estimate.
(3) AIC: Akaike's criterion (5.3).
(4) S: Rao Score tests statistic (5.10).
(5) SL: Significance level.

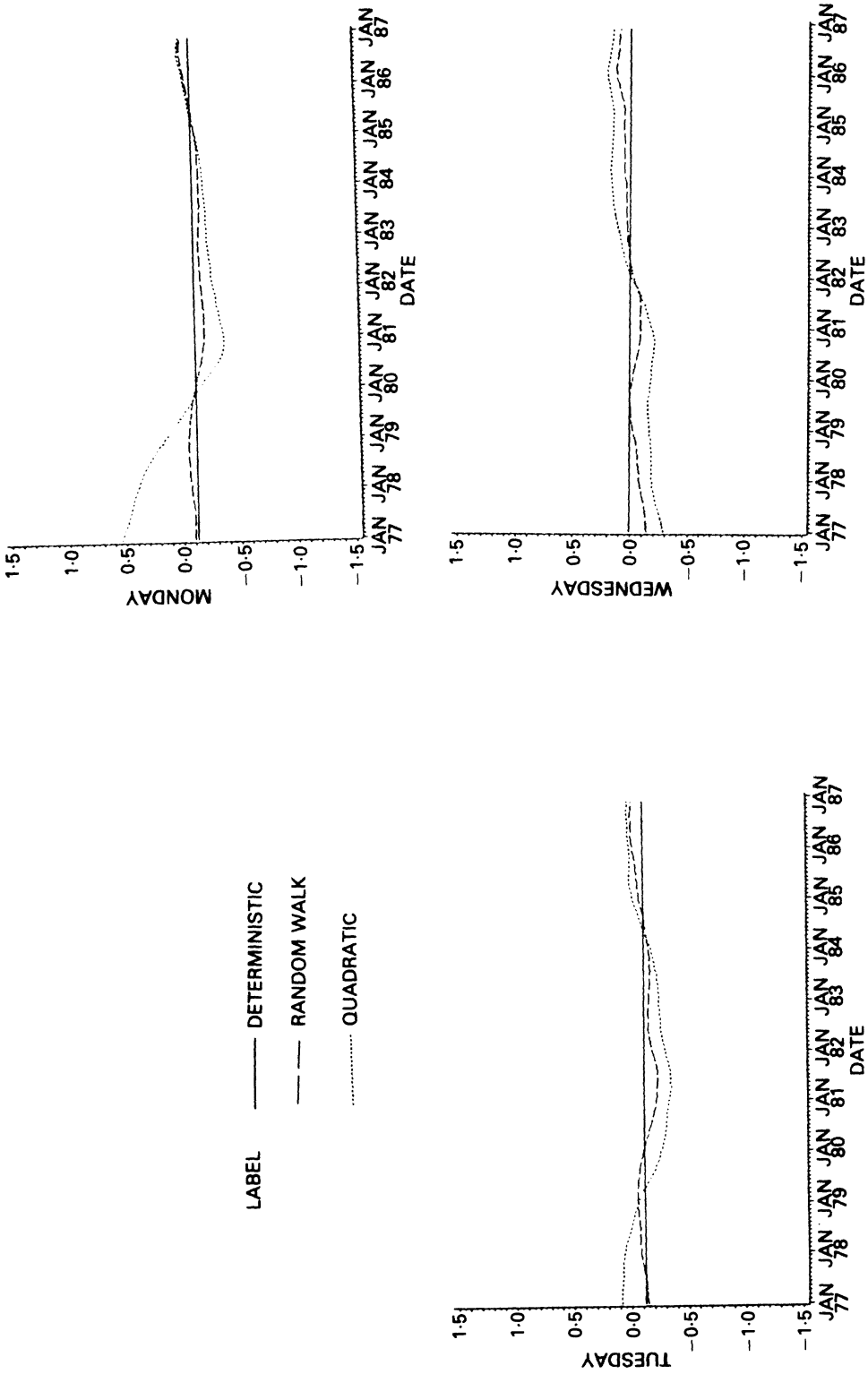


Figure 1A. Total retail trade sales all stores—Nova Scotia. (I)

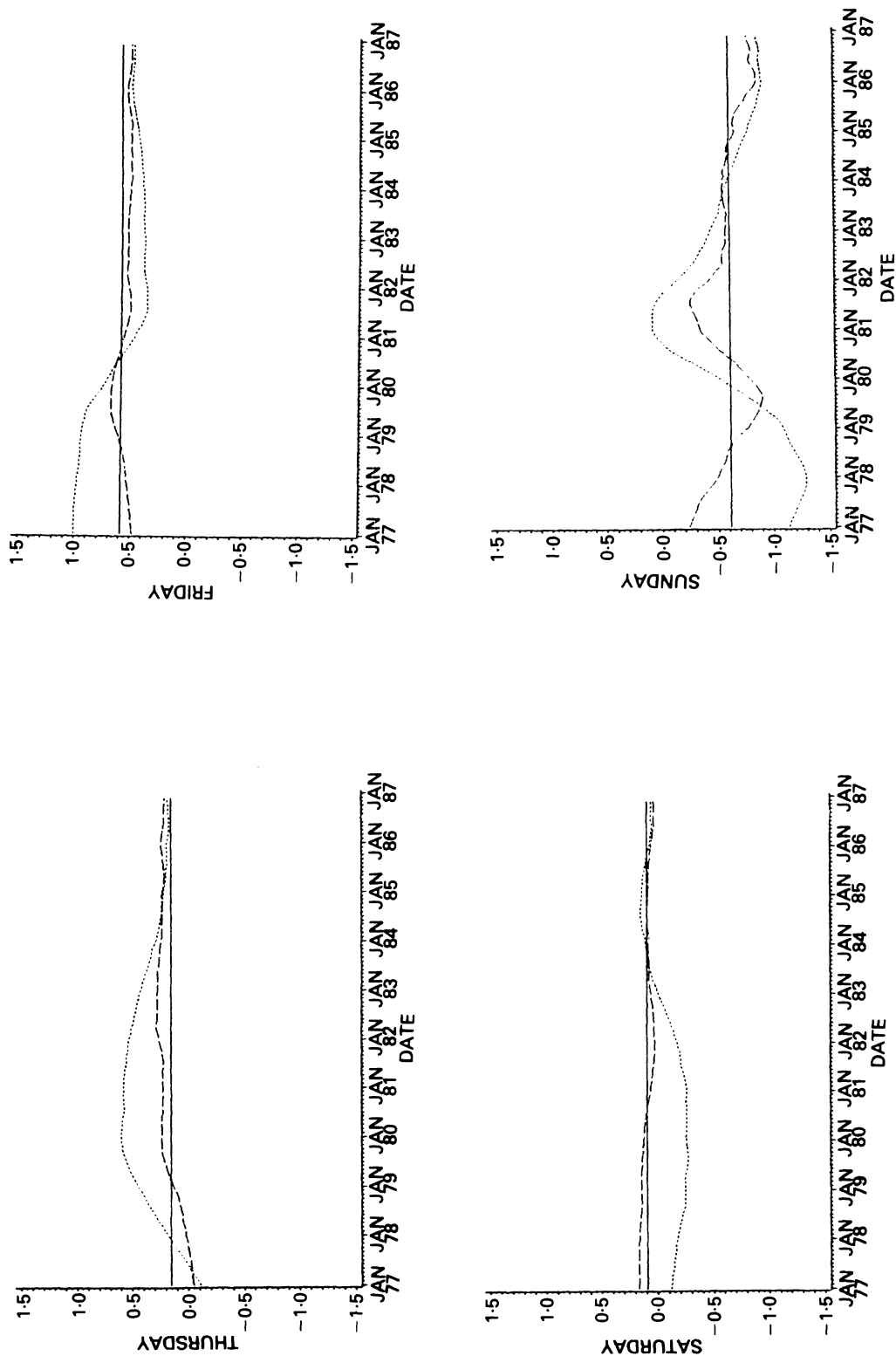


Figure 1A. (Continued)

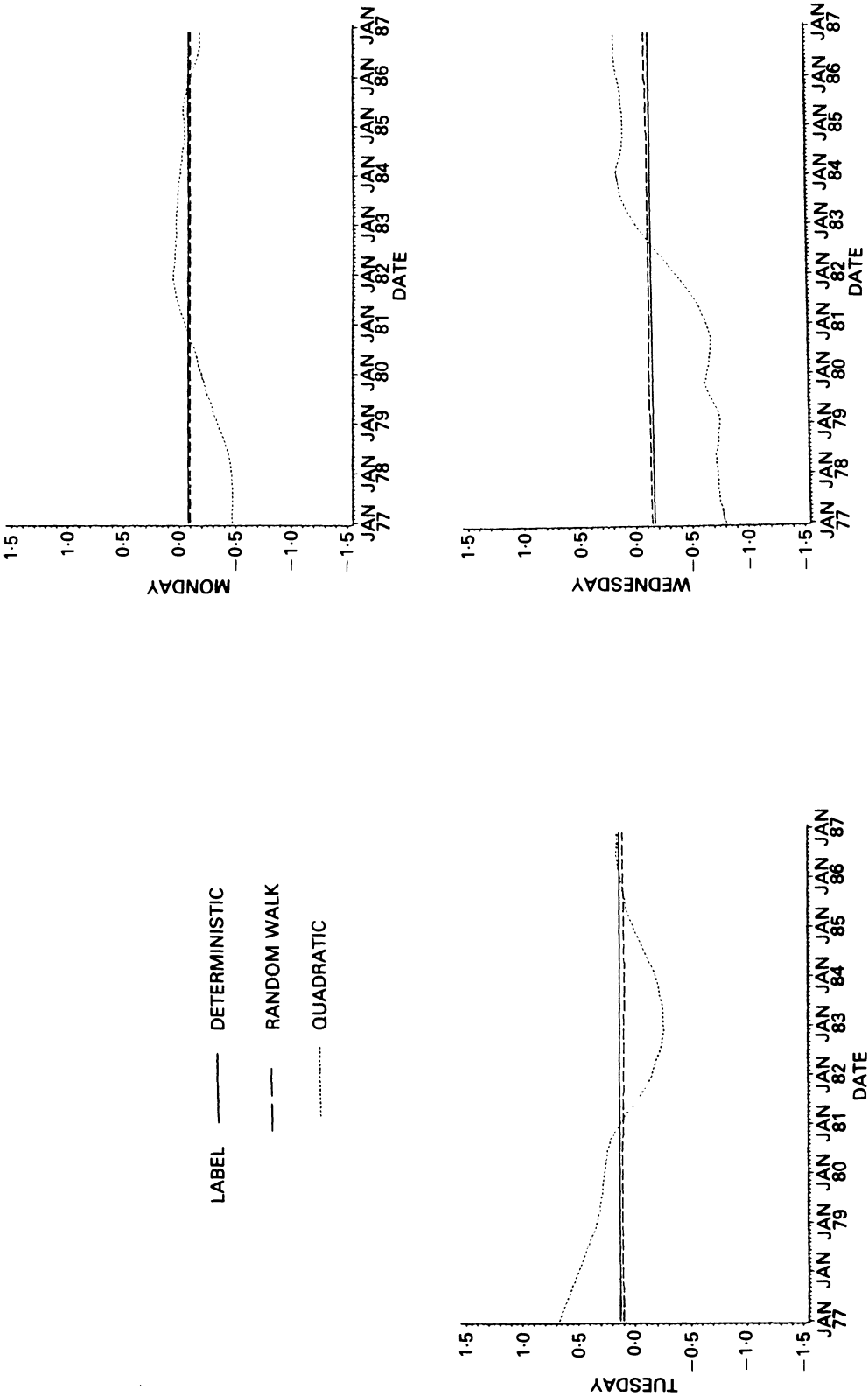


Figure 1B. Total retail trade sales—department stores—Canada. (II)

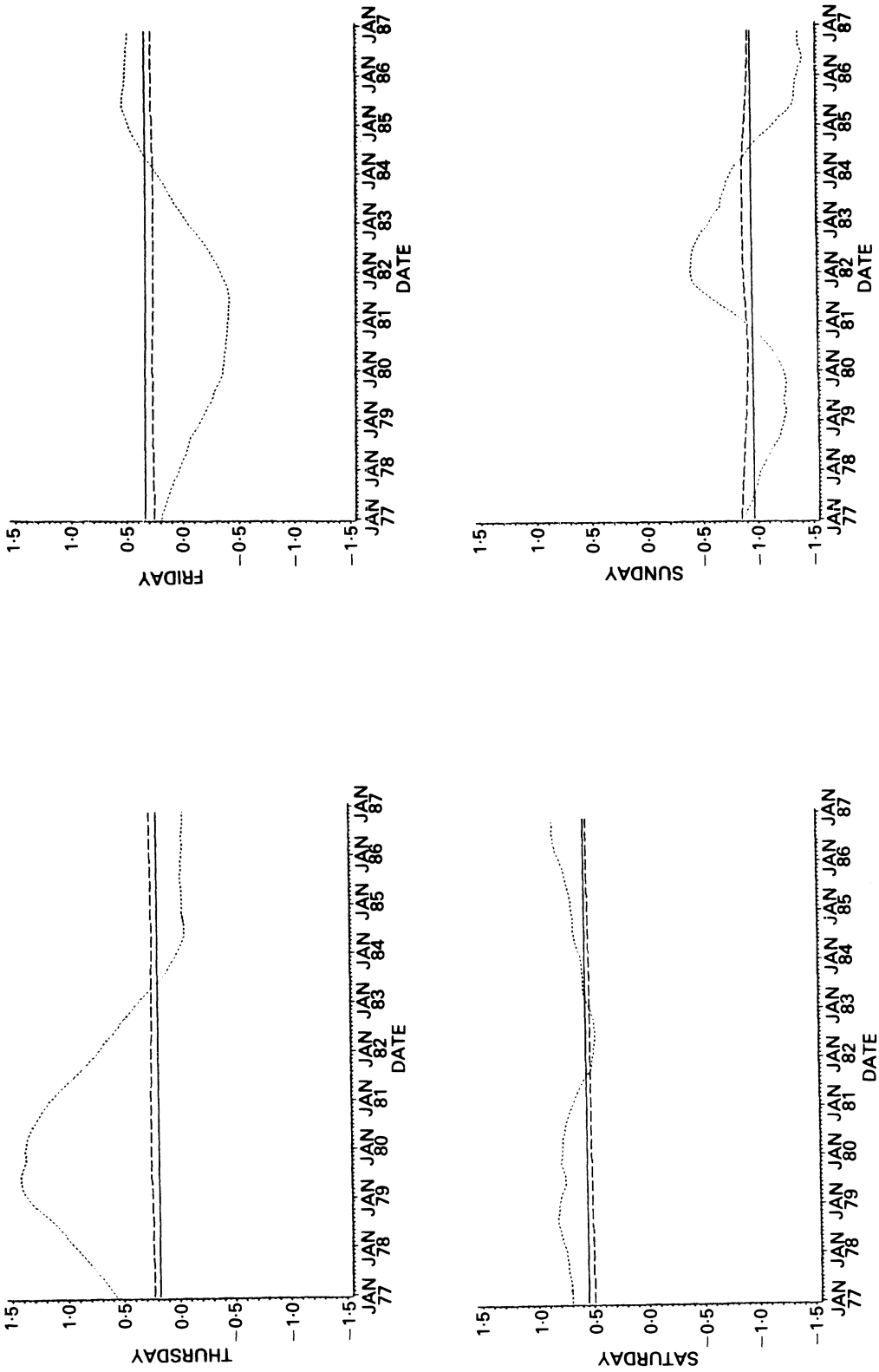


Figure 1B. (Continued)

The estimation of the stochastic trading-day coefficients in a typical application is done iteratively as follows:

- Step1: A first estimation of the parameter σ_x^2/σ^2 is obtained with the Fibonacci line search on the *time-reversed* series given fixed initial conditions $\alpha_0 = 0$ and $\mathbf{P}_0 = k\mathbf{I}$ for the *time-reversed* series ($k = 21$ in the applications given below).
- Step2: The initial conditions α_0 and $\mathbf{P}_0$ are estimated by forecasting them from the time-reversed series with the estimated parameter σ_x^2/σ^2 from step1.
- Step3: A second estimation of the parameter σ_x^2/σ^2 with the Fibonacci line search is done on the original series with the initial conditions from step2.
- Step4: A third and final estimation of the parameter σ_x^2/σ^2 is done with the method of scoring. If the procedure leads to a negative estimate, the initial estimate from step3 is used.
- Step5: The trading-day coefficients are estimated from the original series with the Kalman filter and the fixed interval smoother using the parameter σ_x^2/σ^2 from step4 and the initial conditions from step2.

To illustrate we use two examples with real data. The first example is the series (I) of Total Retail Trade Sales for All Stores in Nova-Scotia from January 1977 to December 1986 and the second example is the series (II) of Total Retail Trade Sales for Department Stores in Canada from January 1977 to December 1986. The input data as obtained from equation (6.1) for both series are given in Tables 1A and 1B.

Table 2 gives a summary of the estimation method for the signal to noise ratio σ_x^2/σ^2 . For both series the $AIC(k)$, and consequently $S(k)$, is smallest with $k = 1$ suggesting that the random walk model is preferred over the quadratic model. The results of the Rao Score test given in (5.10) further suggest that the random walk model is appropriate for the series (I) whereas the deterministic model is more appropriate for the series (II).

Graphs of the trading-day coefficients for both series are provided in Figs. 1A and 1B. Clearly the stochastic model allows for a moving behaviour of the trading-day coefficients. For the series (I), the graphs of the coefficients under the stochastic models have similar paths but the quadratic model ($k = 2$) exaggerates the amplitude. For the series (II), the graphs of the coefficients under the random walk model are almost straight lines. This

Table 3A
Total retail trade sales—all stores—Nova Scotia (I). Spectral densities for the original and trading-day adjusted series for selected frequencies.

freq	Y	Det	Ka ($k = 1$)	Sm ($k = 1$)	Ka ($k = 2$)	Sm ($k = 2$)
0.3278	1.1695	9.1319	6.4164	5.7778	7.3873	7.0939
0.3333	0.6797	3.6927	2.9862	2.3540	4.3460	6.0100
0.3389	5.3868	1.5177	0.9303	1.2743	1.4574	6.1753
0.3444	23.8114	2.0735	2.0121	1.6289	0.7589	4.1351
0.3500	37.6602	6.9724	5.2330	3.5292	3.0434	4.2020
0.3556	22.5998	9.6120	7.1576	5.2519	6.4855	4.7701
0.3611	5.5827	4.8960	5.2180	3.9993	5.8694	2.7787
0.3667	1.4694	1.5090	1.9696	1.7508	2.2158	1.1023
0.3722	1.5816	2.6025	1.2286	1.9509	0.5728	1.6986
sum:	99.9411	42.0077	33.1518	27.5171	32.1365	37.9660

Y: original, Det: Trading-day adjusted by the deterministic model.
Ka ($k = 1$): Trading-day adjusted with the coefficients from the random walk model without smoothing.
Sm ($k = 1$): Same as Ka ($k = 1$) but with smoothing.
Ka ($k = 2$): Same as Ka ($k = 1$) but with the quadratic model ($k = 2$).
Sm ($k = 2$): Same as Sm ($k = 1$) but with $k = 2$.

Table 3B

Total retail trade sales—department stores—Canada (II). Spectral densities for the original and trading-day adjusted series for selected frequencies.

freq	Y	Det	Ka ($k = 1$)	Sm ($k = 1$)	Ka ($k = 2$)	Sm ($k = 2$)
0.3278	0.6511	2.0879	1.7716	1.8740	2.1924	2.9300
0.3333	0.7347	1.1348	0.8986	1.0175	1.6366	3.8936
0.3389	4.5313	1.4962	1.0203	1.3066	3.0590	6.3690
0.3444	15.7876	1.9041	1.4534	1.5555	3.5352	5.4069
0.3500	21.1031	1.4258	1.5010	1.2729	2.0801	3.0818
0.3556	11.6191	1.3611	1.5608	1.5225	1.6078	3.7332
0.3611	5.1573	1.7703	1.7976	1.9902	2.2078	3.6070
0.3667	3.8195	3.2686	3.1939	3.3355	2.5901	2.3057
0.3722	3.0257	5.1427	4.7810	4.9885	2.3659	2.2241
sum:	66.4294	19.5915	18.9782	17.8632	21.2749	33.5513

Y: original, Det: Trading-day adjusted by the deterministic model.

Ka ($k = 1$): Trading-day adjusted with the coefficients from the random walk model without smoothing.

Sm ($k = 1$): Same as Ka ($k = 1$) but with smoothing.

Ka ($k = 2$): Same as Ka ($k = 1$) but with the quadratic model ($k = 2$).

Sm ($k = 2$): Same as Sm ($k = 1$) but with $k = 2$.

agrees with the result from the Rao Score test suggesting that the deterministic model is appropriate for this series.

Next a frequency domain analysis was conducted on both the original series and the trading-day adjusted series which is defined as the original series minus the estimated trading-day variations. Estimates of the trading-day variations from five procedures are compared, namely, the deterministic (Det), the random walk without smoothing (Ka ($k = 1$)), the random walk with smoothing (Sm ($k = 1$)), the quadratic model without smoothing (Ka ($k = 2$)) and the quadratic model with smoothing (Sm ($k = 2$)). Tables 3A and 3B provide the estimates of the spectral densities using the Parzen window for a band of frequencies between 0.3278 and 0.3722. The spectral densities of the input series are characterized by a peak at the frequencies around 0.35. This is typical of series affected by trading-day variations as shown by Cleveland & Devlin (1980). For both series, the largest reduction in power is given by the random walk model with smoothing. However in series (II) the difference between the deterministic and random walk model is very small, in accordance with the result obtained from the Rao Score test.

7 Conclusions

We have introduced two stochastic models for trading-day coefficients and compared them with the deterministic model developed by Young (1965). The stochastic models are given in a state-space form and trading-day coefficients are estimated by the Kalman filter and fixed interval smoother. We have provided solutions to the problems of estimating the parameters and initial conditions of the Kalman filter and to distinguish between alternative models using statistical tests based on the AIC or the Rao Score test statistic. Two real case studies are thoroughly discussed to illustrate when the deterministic and the stochastic models are adequate.

Appendix: Rao Score Test for the Presence of Random Walk Coefficients in Simple Linear Model

Let $\mathbf{P}$ satisfy $\mathbf{P}'\mathbf{P} = \mathbf{I}_{T-6}$, $\mathbf{P}\mathbf{P}' = (\mathbf{I}_T - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}') = \mathbf{M}$ and $\mathbf{P}'\mathbf{X} = \mathbf{0}$. Rao Score test is given by,

$$S = \frac{U^2(\sigma_x^2 = 0, \hat{\sigma}_0)}{I(\sigma_x^2 = 0, \hat{\sigma}_0)}$$

where

$$U(\sigma_x^2 = 0, \hat{\sigma}_0) = \frac{\partial}{\partial \sigma_x^2} \text{Log } L \Big|_{\sigma_x^2=0, \sigma^2=\hat{\sigma}_0^2},$$

$$I(\sigma_x^2 = 0, \hat{\sigma}_0) = \mathcal{E} \left(\frac{-\partial^2}{\partial (\sigma_x^2)^2} \log L \right) \Big|_{\sigma_x^2=0, \sigma^2=\hat{\sigma}_0^2},$$

and $\text{Log } L$ is the log-likelihood of the observations given by,

$$\text{Log } L = -\frac{1}{2} \log |\Sigma| - \frac{1}{2} \mathbf{z}' \Sigma^{-1} \mathbf{z}.$$

Using the following results:

$$\frac{\partial}{\partial \sigma^2} \Sigma = \mathbf{I}_{T-6}, \quad \frac{\partial}{\partial \sigma_x^2} \Sigma = \mathbf{P}' \mathbf{V} \mathbf{P},$$

$$\frac{\partial}{\partial x} \text{Log } |\Sigma| = \text{trace} \left(\Sigma^{-1} \frac{\partial}{\partial x} \Sigma \right), \quad \frac{\partial}{\partial x} \Sigma^{-1} = -\Sigma^{-1} \left(\frac{\partial}{\partial x} \Sigma \right) \Sigma^{-1},$$

we can deduce,

$$\frac{\partial}{\partial \sigma_x^2} \text{Log } L = -\frac{1}{2} \text{trace} (\Sigma^{-1} \mathbf{P}' \mathbf{V} \mathbf{P}) + \frac{1}{2} \mathbf{z}' \Sigma^{-1} \mathbf{P}' \mathbf{V} \mathbf{P} \Sigma^{-1} \mathbf{z},$$

$$\frac{\partial^2}{\partial (\sigma_x^2)^2} \text{Log } L = \frac{1}{2} \text{trace} (\Sigma^{-1} \mathbf{P}' \mathbf{V} \mathbf{P} \Sigma^{-1} \mathbf{P}' \mathbf{V} \mathbf{P}) - \mathbf{z}' \Sigma^{-1} \mathbf{P}' \mathbf{V} \mathbf{P} \Sigma^{-1} \mathbf{P}' \mathbf{V} \mathbf{P} \Sigma^{-1} \mathbf{z},$$

$$\mathcal{E} \left(-\frac{\partial^2}{\partial (\sigma_x^2)^2} \text{Log } L \right) = \frac{1}{2} \text{trace} (\Sigma^{-1} \mathbf{P}' \mathbf{V} \mathbf{P} \Sigma^{-1} \mathbf{P}' \mathbf{V} \mathbf{P}).$$

Under H_0 : $\Sigma = \sigma^2 \mathbf{I}_{T-6}$, $\Sigma^{-1} = \sigma^{-2} \mathbf{I}_{T-6}$ and the model for the observations is $\mathbf{z} \sim N(0, \sigma^2 \mathbf{I}_{T-6})$. It follows that the MLE of σ^2 under H_0 is,

$$\hat{\sigma}_0^2 = \frac{1}{T-6} \mathbf{z}' \mathbf{z} = \frac{1}{T-6} \mathbf{w}' \mathbf{P} \mathbf{P}' \mathbf{w} = \frac{1}{T-6} \mathbf{w}' \mathbf{M} \mathbf{w}$$

i.e., the usual unbiased estimate of σ^2 under the deterministic model. Under H_0 the score is,

$$U(\sigma_x^2 = 0, \hat{\sigma}_0^2) = \frac{1}{2 \hat{\sigma}_0^4} \mathbf{z}' \mathbf{P}' \mathbf{V} \mathbf{P} \mathbf{z} - \frac{1}{2 \hat{\sigma}_0^2} \text{trace} (\mathbf{P}' \mathbf{V} \mathbf{P})$$

$$= \frac{1}{2 \hat{\sigma}_0^4} \hat{\mathbf{e}}' \mathbf{V} \hat{\mathbf{e}} - \frac{1}{2 \hat{\sigma}_0^2} \text{trace} (\mathbf{M} \mathbf{V})$$

where $\hat{\mathbf{e}}$ is the residual vector under the deterministic model and the information is:

$$I(\sigma_x^2 = 0, \hat{\sigma}_0^2) = \frac{1}{2 \hat{\sigma}_0^4} \text{trace} (\mathbf{P}' \mathbf{V} \mathbf{P} \mathbf{P}' \mathbf{V} \mathbf{P})$$

$$= \frac{1}{2 \hat{\sigma}_0^4} \text{trace} (\mathbf{M} \mathbf{V} \mathbf{M} \mathbf{V}).$$

The score statistic is therefore given in equation (5.10).

To test the significance level of the S statistic we did a small simulation study. We used the deterministic trading-day coefficients estimated from series (II) as well as the estimate of σ^2 to simulate 2000 series of the same length (10 years) corresponding to the same X -matrix. For each simulated series the value of the S statistic and its corresponding

significance level from a chi-square one distribution were calculated. The results showed that the estimated 5% significance level was 0.047.

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Résumé

Un grand nombre de séries de flux et de stock reliés à la production, ventes et inventaires sont affectées par la rotation des jours ou l'effet du calendrier. La composante de rotation des jours représente les variations intra-mensuelles causées par la distribution des jours dans le mois (ex: 4 lundis, ... 4 vendredis, 5 samedis et 5 dimanches dans un mois de 30 jours). Ces variations sont systématiques et peuvent fortement influencer les comparaisons entre les mois. Ainsi, lorsque ces variations sont présentes dans une série, elles sont généralement soustraites avec la saisonnalité et possiblement d'autres effets, tel que l'effet de Pâques, pour obtenir un estimé plus précis de la tendance-cycle.

Un modèle de régression multiple avec coefficients fixes, développé par Young (1965), est utilisé par les méthodes X-11-ARMMI (Dagum, 1980) et X11 (Shiskin, Young et Musgrave, 1967) pour l'estimation de la composante de rotation des jours. Ce modèle assume que les poids des jours et leur patron hebdomadaire restent constants à travers une période pré-déterminée de la série. Pour certaines séries socio-économiques cette hypothèse peut être discutable.

Cet article présente un modèle de régression multiple avec coefficients aléatoires pour l'estimation de la composante de rotation des jours. Les méthodes d'estimation et d'inférence sont discutées et les modèles déterministe et aléatoire sont appliqués à des séries réelles.

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